Automatic clothing category recognition

Phase 1 / Project 11 / Section A

Inventory image tagging with pixels and HOG

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Research question and prior work

Does local edge structure improve clothing classification?

Raw pixels

Intensity at each fixed image position. A simple, direct baseline.

HOG descriptors

Histograms of local edge directions. The representation stays fixed during classifier training.

Fashion-MNIST: Xiao et al. (2017) HOG: Dalal & Triggs (2005)

Team responsibilities

Member	Responsibility
Nakul	Integration, literature and reporting
Sudeshhna	Data quality, EDA and shared splits
Puneet	Pixels, HOG, Logistic Regression and kNN
Samarth	Random Forest, SVM and experiment management
Neelanshu	Evaluation, timing and error analysis

Fashion-MNIST dataset

60,000 training images **10,000 test images** **28 × 28 grayscale**



T-shirt/top



Trouser



Pullover



Dress



Coat



Sandal



Shirt



Sneaker



Bag



Ankle boot

Official Zalando Fashion-MNIST. Seeded examples from the training fit pool.

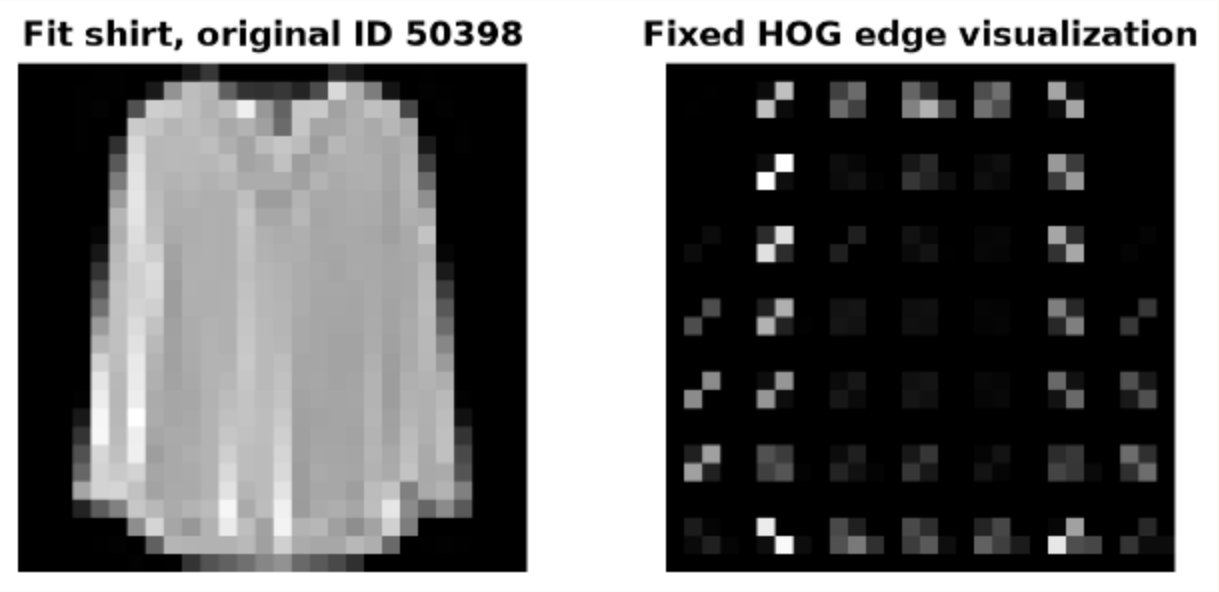
Data checks and fair partitions

Partition	Images	Use
Fit pool	48,000	Fit all eight final models
Validation	12,000	Choose settings and overall model
Official test	10,000	Evaluate after selection freeze

Seed 42, stratified membership, saved original IDs

Checks passed: shape, range, labels, exact duplicates and split overlap.
Near-duplicate detection and a full semantic label audit remain outside scope.

Pixels and HOG features



Pixels: 28 × 28 = 784

HOG: 6 × 6 × 2 × 2 × 9
= 1,296 features

HOG settings: 9 orientations, 4 × 4 cells, 2 × 2 blocks, L2-Hys
Normalize pixels to [0, 1]. Fit any StandardScaler on the fit pool only.

Logistic Regression and kNN

Logistic Regression

Learns linear category scores. C controls regularization strength.

k nearest neighbors

Uses distance-weighted votes from stored examples. k controls the neighborhood.

Pipeline	Preprocessing
LR with pixels or HOG	Training-only StandardScaler
kNN with pixels	Pixel normalization only
kNN with HOG	Training-only StandardScaler

Random Forest and RBF SVM

Random Forest

200 randomized decision trees. Depth controls rule complexity. No StandardScaler.

RBF SVM

Nonlinear similarity and margin decisions. StandardScaler, gamma = scale, no probability estimation.

The same fit and validation IDs support all four model families

Model selection before test access

Stage	Images used	Decision
Smoke check	2,000	Verify the pipeline
Compact development search	12,000	Two candidates per feature/model row
Final fitting	48,000	Fit eight selected configurations
Freeze, then official test	10,000 test	Evaluate the fixed finalists

Primary selection metric: validation macro F1

All eight final models share the same 48,000 fit images.

Evaluation measures

Accuracy

Correct predictions / all images

Macro F1

Mean of the ten class F1 scores. Each category has equal weight.

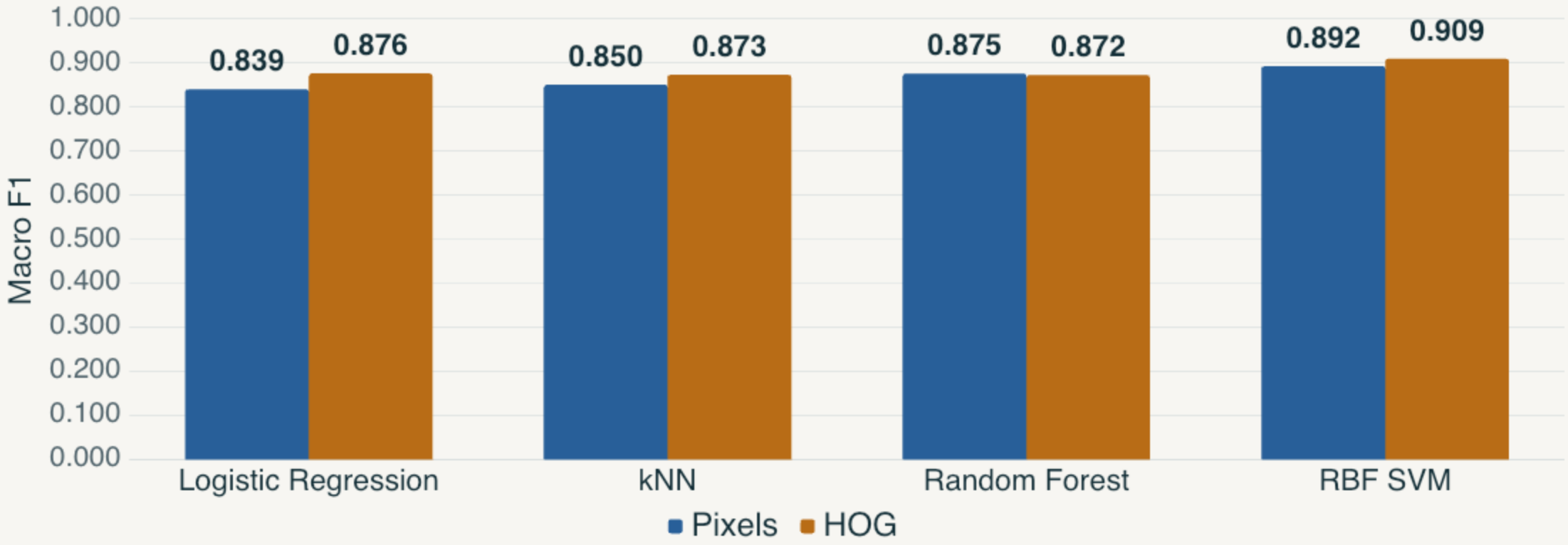
Per-class recall

Correctly identified members of a class / all true members of that class

Operational cost

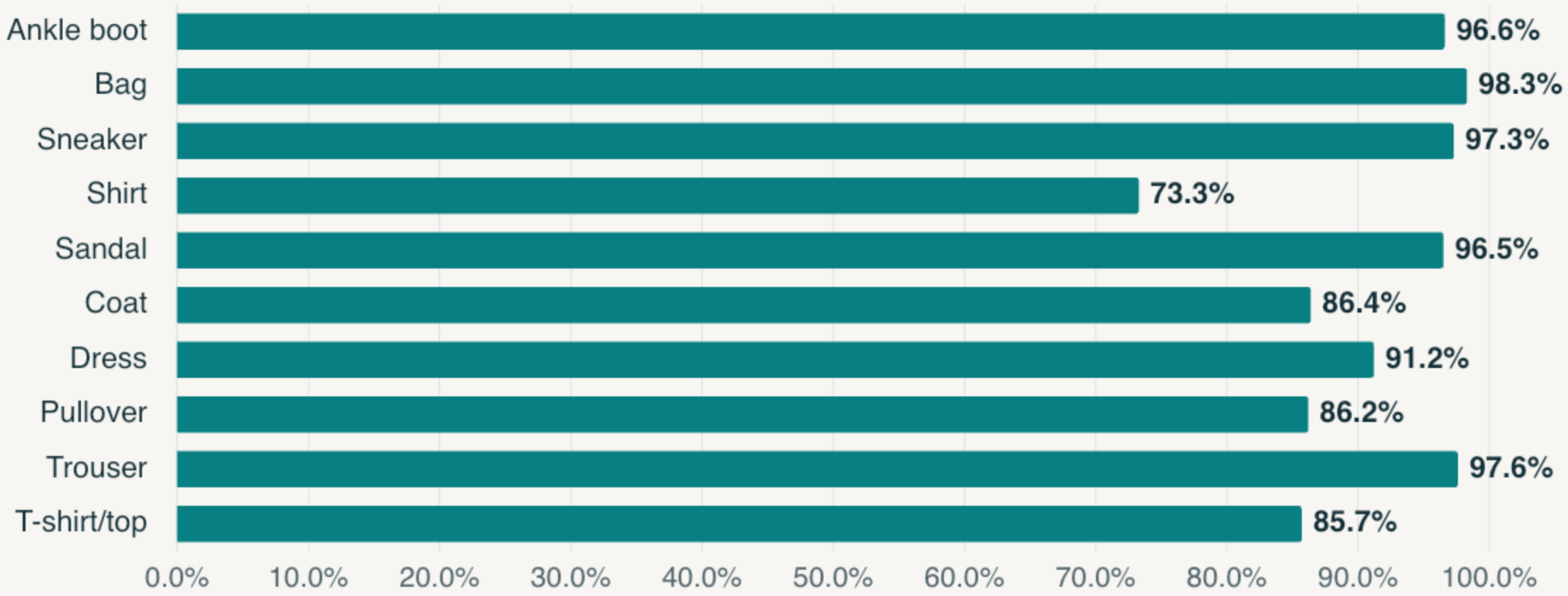
Repeated inference measurements and model-specific storage quantities

HOG helps three classifier families



Selected HOG + RBF SVM: 90.91% accuracy, 0.9090 macro F1

Shirt recall remains the weakest



Selected model. Official test: 1,000 images per category.

Confusion among upper-body categories

Actual / predicted	T-shirt/top	Pullover	Coat	Shirt	Other
T-shirt/top	857	14	4	104	21
Pullover	12	862	57	59	10
Coat	3	51	864	42	40
Shirt	103	63	64	733	37

Rows: true labels. Columns: predictions. Each row totals 1,000.

Other combines the six remaining predicted categories.

Paired SVM prediction examples

Only HOG correct
Test ID 17



True: Coat
Pixels: Pullover
HOG: Coat

Only HOG correct
Test ID 49



True: Pullover
Pixels: Shirt
HOG: Pullover

Both wrong
Test ID 12



True: Sneaker
Pixels: Sandal
HOG: Sandal

Both wrong
Test ID 23



True: Ankle boot
Pixels: Sandal
HOG: Sandal

First sample IDs per outcome group. Examples illustrate errors, not prevalence.

Inference cost changes the trade-off

Classifier	Pixels (ms/image)	HOG (ms/image)
Logistic Regression	0.0014	0.1473
kNN	0.3298	0.9880
Random Forest	0.0609	0.2061
RBF SVM	3.2468	21.7090

Median end-to-end cost, batch 128, 5 repeats, 1 native thread

Includes feature extraction. Excludes loading. One machine and 1,024 fixed images.

Reproducible Phase 1 evidence

Trace each result

Saved IDs, candidate logs, prediction files and a hashed freeze manifest

Check the calculations

Executed notebook, independent metric audit and eight integrity tests

github.com/Nakul-Jaglan/aml-dl-t089

Source, results, figures and report share the same frozen experiment evidence.

Conclusions and Phase 2 proposal

90.91%

Test accuracy of HOG + RBF SVM

Shirt recall: 73.3%. Higher accuracy carries a larger inference cost.

Limits: one split, compact search, fixed preprocessing and 28 × 28 images.

Phase 2: test a small CNN on the same saved partitions.
CNN performance is unmeasured.

Selected references

Xiao et al. (2017) Fashion-MNIST
arxiv.org/abs/1708.07747

Dalal & Triggs (2005) Histograms of Oriented Gradients
doi.org/10.1109/CVPR.2005.177

Breiman (2001) Random Forests
doi.org/10.1023/A:1010933404324

Cortes & Vapnik (1995), Support-vector networks
Pedregosa et al. (2011), Scikit-learn: Machine Learning in Python
LeCun et al. (1998), Gradient-Based Learning Applied to Document Recognition

All eight finalists

Pipeline	Val. F1	Test acc.	Test F1
Pixels + Logistic Regression	0.8590	84.01%	0.8394
Pixels + kNN	0.8585	85.04%	0.8505
Pixels + Random Forest	0.8828	87.67%	0.8753
Pixels + RBF SVM	0.9068	89.28%	0.8924
HOG + Logistic Regression	0.8834	87.60%	0.8757
HOG + kNN	0.8749	87.34%	0.8730
HOG + Random Forest	0.8752	87.29%	0.8718
HOG + RBF SVM	0.9114	90.91%	0.9090

Selection used validation F1. The official test did not choose the winner.

Selected settings and search scope

Pipeline	Selected setting	Scaling
Pixels + Logistic Regression	C = 0.1	StandardScaler
Pixels + kNN	k = 5	None
Pixels + Random Forest	depth = None	None
Pixels + RBF SVM	C = 10	StandardScaler
HOG + Logistic Regression	C = 0.1	StandardScaler
HOG + kNN	k = 5	StandardScaler
HOG + Random Forest	depth = None	None
HOG + RBF SVM	C = 10	StandardScaler

Two candidates per feature/model pair. Fixed RF: 200 trees. Fixed SVM: RBF.

Model size and complexity

Pipeline	Model-specific quantity	MiB
Pixels + Logistic Regression	7,850 weights + bias	0.05
Pixels + kNN	37,632,000 stored feature scalars	31.10
Pixels + Random Forest	1,633,648 tree nodes	35.01
Pixels + RBF SVM	13,725,289 decision-array scalars	39.34
HOG + Logistic Regression	12,970 weights + bias	0.08
HOG + kNN	62,208,000 stored feature scalars	147.27
HOG + Random Forest	1,828,070 tree nodes	40.24
HOG + RBF SVM	27,295,425 decision-array scalars	87.82

Counts differ by model family. MiB is serialized size, not peak memory.

Viva and saved-prediction demo

Core questions

Why keep validation separate?
Why is HOG 1,296-dimensional?
Why can scaling change kNN?

Offline demonstration

Open the executed notebook.
Inspect original test ID 0.
Recompute a saved metric.

Explain one error and one limitation in your own words